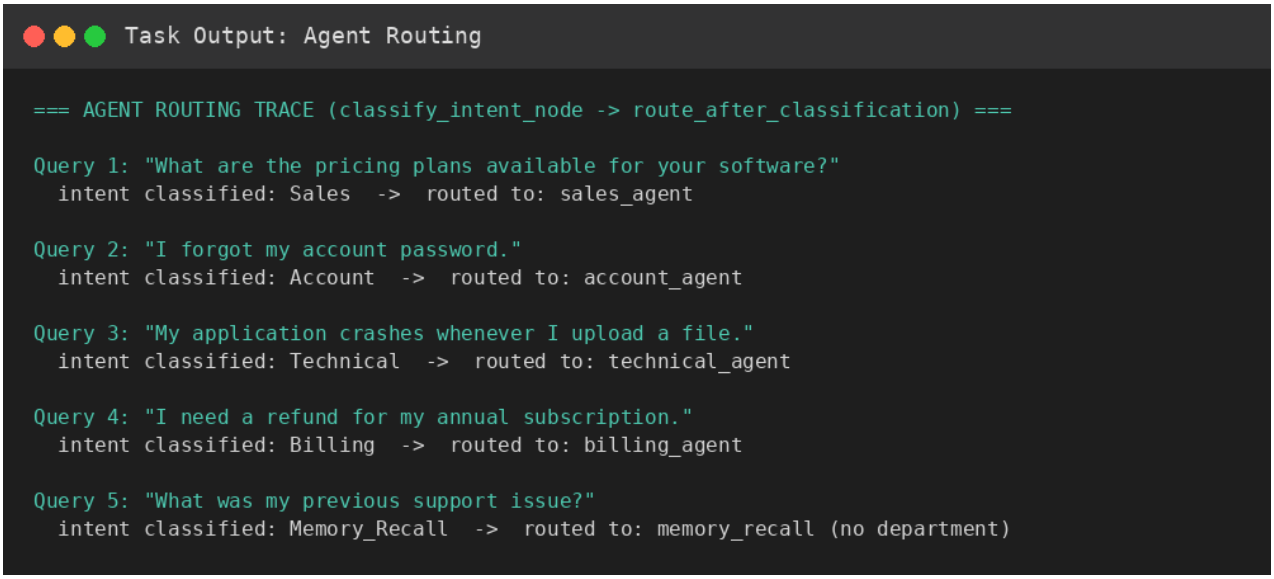
AI-Powered Customer Support Automation System

Task Output Screenshots

Agent Routing | RAG Retrieval | Human-in-the-Loop | Memory Storage & Recall | Final Response Generation

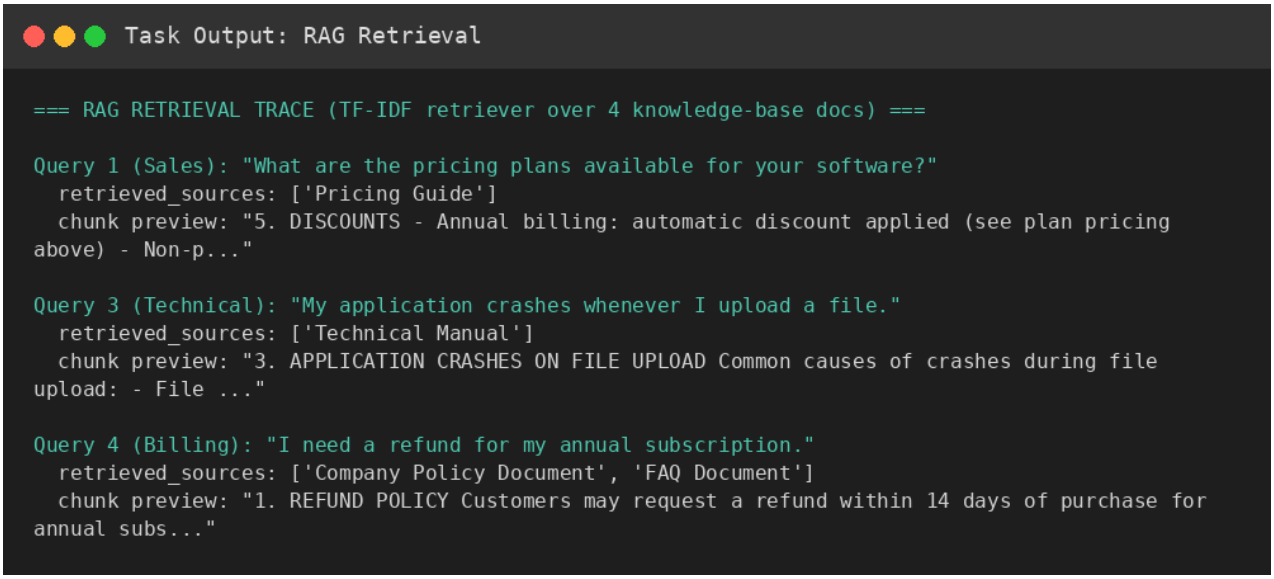
1. Agent Routing

Demonstrates classify_intent_node correctly classifying each of the five sample queries and route_after_classification sending them to the matching department agent.



2. RAG Retrieval

Demonstrates the TF-IDF retriever pulling the most relevant knowledge-base chunk for each department (Pricing Guide for Sales, Technical Manual for Technical Support, Company Policy / FAQ for Billing).



3. Human-in-the-Loop

Demonstrates the refund request triggering LangGraph's interrupt(), pausing graph execution, and resuming correctly once a supervisor decision is supplied via Command(resume=...).

Task Output: Human-in-the-Loop Approval

```
=== HUMAN-IN-THE-LOOP APPROVAL TRACE ===

Query 4: "I need a refund for my annual subscription."

>>> graph.invoke() returned with __interrupt__ <<<
  risk_reason: refund_request
  customer_id: cust_demo_david
  draft_response (awaiting approval): "Thank you for reaching out to Billing Support. Based on our
records: - 1. REFUND POLICY Regarding yo..."

Graph execution PAUSED. Waiting for human supervisor decision...

Supervisor decision submitted via Command(resume=...):
  decision: approve
  notes: "Approved per policy: within 14-day refund window."

>>> Graph resumed <<<
  approval_status: approved
  supervisor_notes: Approved per policy: within 14-day refund window.
```

4. Memory Storage & Recall

Demonstrates every turn being persisted to the SQLite conversation_history table, and the Memory_Recall query correctly retrieving the customer's actual previous issue.

Task Output: Memory Storage & Recall

```
=== SQLITE MEMORY: STORAGE & RECALL TRACE ===

Stored conversation_history rows for customer_id='cust_demo_david':

  turn_id=1 | intent=Billing | dept=Billing | query="My name is David. I have a billing issue with
my l..."
  turn_id=2 | intent=Sales | dept=Sales | query="What are the pricing plans available for your
soft..."
  turn_id=3 | intent=Account | dept=Account | query="I forgot my account password...."
  turn_id=4 | intent=Technical | dept=Technical Support | query="My application crashes whenever I
upload a file...."
  turn_id=5 | intent=Billing | dept=Billing | query="I need a refund for my annual
subscription...."
  turn_id=6 | intent=Memory_Recall | dept=Memory | query="What was my previous support issue?..."

Query 5: "What was my previous support issue?"
  intent classified: Memory_Recall  (Memory_Recall, bypasses department routing)
  memory.get_last_issue('cust_demo_david') -> turn_id=5, department=Billing

  FINAL RESPONSE TO CUSTOMER:
  "Your previous issue was related to Billing support. You asked: "I need a refund for my annual
subscription.". Our response at the time was: "Thank you for reaching out to Billing Support.
Based on our records: - 1. REFUND POLICY Regarding your message — "I need a refund for my annual
subscription." "
```

5. Final Response Generation

Demonstrates the final, validated response sent to the customer for each of the five sample queries, including the human-approved refund response.

Task Output: Final Response Generation

```
=== FINAL RESPONSE GENERATION (after Supervisor QA / Human Approval) ===

Query 1 (Sales):
  approval_status: not_required
  final_response: "Thank you for reaching out to Sales Support. Based on our records: - 5.
DISCOUNTS Regarding your message — "What are the pricing plans avail..."

Query 2 (Account):
  approval_status: not_required
  final_response: "Thank you for reaching out to Account Support. Based on our records: - Q: How
do I reset my password? Regarding your message — "I forgot my ..."

Query 3 (Technical):
  approval_status: not_required
  final_response: "Thank you for reaching out to Technical Support. Based on our records: - 3.
APPLICATION CRASHES ON FILE UPLOAD Regarding your message — "My ..."

Query 4 (Billing - human-approved):
  approval_status: approved
  final_response: "Thank you for reaching out to Billing Support. Based on our records: - 1.
REFUND POLICY Regarding your message — "I need a refund for my ann..."

Query 5 (Memory Recall):
  approval_status: not_required
  final_response: "Your previous issue was related to Billing support. You asked: "I need a refund
for my annual subscription.". Our response at the time was: ..."
```